DỰ ÁN CÔNG NGHỆ THÔNG TIN 2

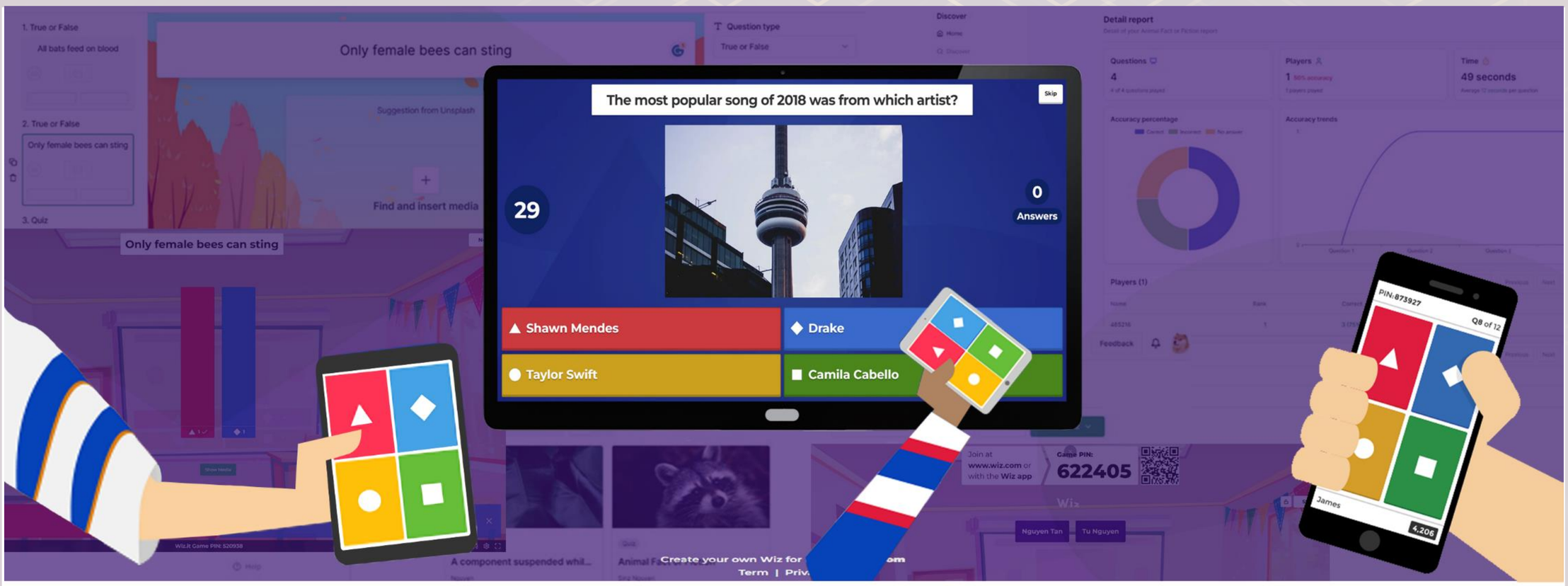
Năm triển khai: 2023

Đề tài: Phát triển ứng dụng thi trắc nghiệm online tương tự Kahoot theo hướng Microservices và Event Driven Architecture

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Xem chi tiết



ABSTRACT

Our website utilizing gamification to provide a multiple-choice question-based learning, demonstrating the potential as digital learning aids. Gamification is frequently used to increase the incentive to use digital learning tools. Typically, the effect of gamification alone is insufficient to create long-term self-directed engagement of students with digital learning tools. In alongside making learning moments more enjoyable. With an exceedingly diversified quiz creation tool, our website allows users to express their creativity.

OBJECTIVE

Our objective is to enhance user experience by automating quizzes, reducing manual labor, and facilitating administrative operations for host-role users.

USE CASE DIAGRAM

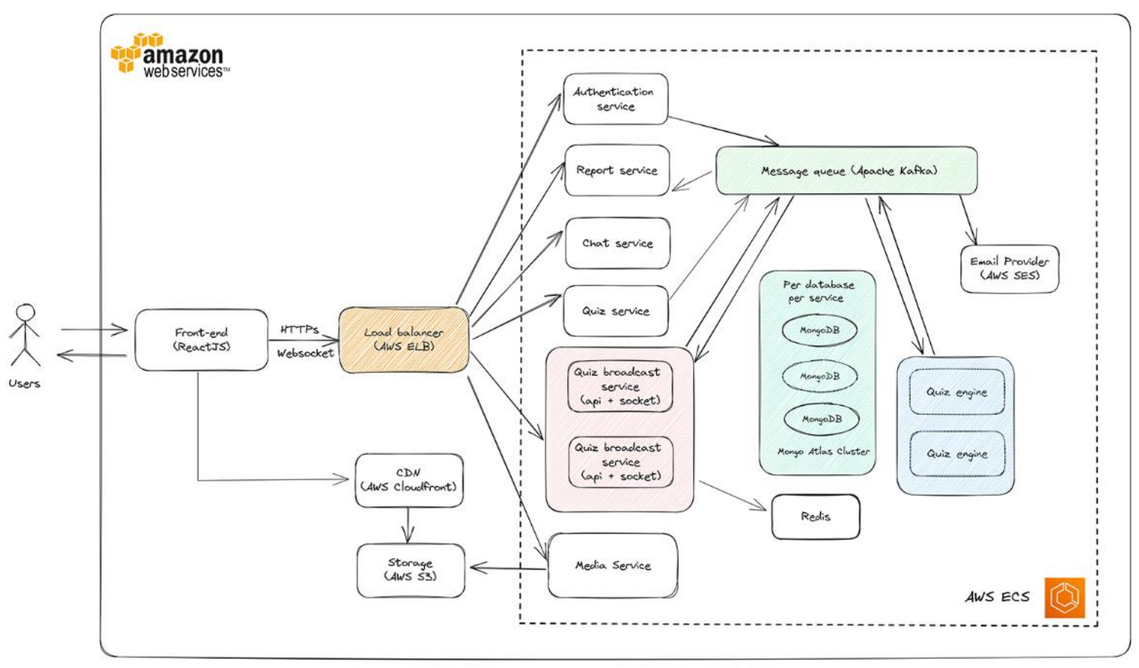
TECHNOLOGIES

For our quiz application, we have 8 services built using Node.js. Each microservice is connected to a set of replica MongoDB databases. Apache Kafka will be used as a message queue to facilitate communication among our microservices for distributed and scalable purposes.

Our services are deployed within a private network on AWS ECS. We utilize AWS ELB (Load balancer) to automatically distribute incoming traffic across multiple targets, thereby enhancing application availability and scalability. Additionally, we make use of Amazon S3 (Storage) and Amazon CloudFront (CDN) to protect and efficiently transport static resources.

In addition, we have implemented Terraform to orchestrate the provisioning of infrastructure and SonarQube also used to ensure code quality. This integration seamlessly aligns with our robust CI/CD pipeline.

ARCHITECTURE



KHOA CÔNG NGHỆ THÔNG TIN
TRƯỜNG ĐẠI HỌC TÔN ĐỨC THẮNG